9.2 Special Right Triangles

Name _____ Period ____ Date _____

Geometry 9.2 · Student sheet

I can…

- find the side lengths of a 45°-45°-90° triangle from one side.
- find the side lengths of a 30°-60°-90° triangle from one side.
- use special right triangles to solve real-life problems.

Vocabulary · fill in the missing word

45°-45°-90° triangle — an _____ right triangle; its two legs are congruent

30°-60°-90° triangle — a right triangle with _____ of 30°, 60°, and 90°; half of an equilateral triangle

Shorter leg — in a 30°-60°-90° _____, the leg across from the 30° angle

Part 1 · Discovery: fold the special triangles

Square: fold it on a diagonal. The right triangle has angles 45°, 45°, 90°.

Leg	Hypotenuse	Hypotenuse ÷ leg

Equilateral triangle: fold it in half along an altitude. The right triangle has angles 30°, 60°, 90°.

Shorter leg	Longer leg	Hypotenuse	Hypotenuse ÷ shorter leg	Longer leg ÷ shorter leg

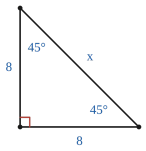
Check on a calculator: $\sqrt{2} \approx$ _____ and $\sqrt{3} \approx$ _____

Part 2 · Notes and examples (fill in each note, then work the example)

45°-45°-90° Triangle Theorem: the hypotenuse is _____ times as long as each leg: hypotenuse = leg · $\sqrt{2}$.

Example 1

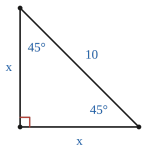
Find the length of the hypotenuse.



From the hypotenuse: to find a leg, _____ the hypotenuse by $\sqrt{2}$, then rationalize.

Example 2

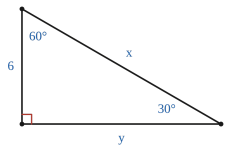
The hypotenuse of a 45°-45°-90° triangle is 10. Find the length of each leg.



30°-60°-90° Triangle Theorem: the hypotenuse is _____ times the shorter leg, and the longer leg is _____ times the shorter leg.

Example 3

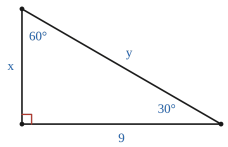
The shorter leg of a 30°-60°-90° triangle is 6. Find the hypotenuse and the longer leg.



Shorter leg first: in a 30°-60°-90° triangle, always find the _____ leg first.

Example 4

The longer leg of a 30°-60°-90° triangle is 9. Find the shorter leg and the hypotenuse.



Example 5

A road sign is an equilateral triangle with sides of 18 inches. How tall is the sign? Round to the nearest tenth.

Part 3 · Partner practice

Partner 1

The hypotenuse of a 45°-45°-90° triangle is $6\sqrt{2}$. Find the length of each leg.

Partner 2

The hypotenuse of a 30°-60°-90° triangle is 14. Find the shorter leg and the longer leg.